Metric Spaces & Topology

An exam-focused study guide to Chapters 0–5 for your Tuesday 10 November session (2:00–7:00pm)

Ch 0 Preliminaries | Ch 1 Metric spaces | Ch 2 Open & closed sets, convergence, topologies | Ch 3 Continuity & homeomorphisms | Ch 4 Completeness | Ch 5 Compactness

HOW TO USE THIS GUIDE

Each chapter starts with a **time slot** from your plan and a short list of what to do in it. Inside each chapter you'll find:

- **Intuition**: the idea in plain English before any symbols.
- **Definitions** (blue): the exact wording to memorise for exams.
- **Theorems** (purple), with proofs of the results that are most often examined.
- **Examples** (green) and **common mistakes** (orange).
- **Practice questions** marked easy, medium or hard, each with a full solution. Try each one before you open the solution.

At the end there's a **proof-template toolkit**, a **counterexample bank**, a **glossary** of every term, and the **recap checklist** for your 5:35 slot.

Chapter numbering follows the usual order for this course. If your notes order the topics differently, follow the topic headings rather than the chapter numbers.

Your session plan: Tuesday 10 November

Time	Task	Section of this guide
2:00–2:10	Warm-up: skim the headings of every chapter	Contents below
2:10–2:30	Ch 0: quick review of prerequisites	Chapter 0
2:30–3:05	Ch 1: metric spaces	Chapter 1
3:05–3:15	☕ Break	
3:15–3:50	Ch 2: open/closed sets, convergence, topologies	Chapter 2
3:50–4:25	Ch 3: continuity	Chapter 3
4:25–4:40	☕ Longer break	
4:40–5:10	Ch 4: completeness	Chapter 4
5:10–5:35	Ch 5: compactness	Chapter 5
5:35–5:45	Topology recap from memory	Recap checklist
5:45–6:00	☕ Break	
6:00–6:50	Linear systems	Separate guide (to follow)
6:50–7:00	Wrap-up: what's done and where to start next time	

Contents

0. Preliminaries: sets, functions, $\mathbb{R}$, sequences
1. Metric spaces
2. Open and closed sets, convergence, topological spaces
3. Continuity and homeomorphisms
4. Completeness and the Contraction Mapping Theorem
5. Compactness

➕ Bonus: connectedness (if your course covers it)

🧰 Proof-template toolkit

🧪 Counterexample bank

📖 Glossary of all terminology

✅ 5:35 recap checklist

NOTATION USED THROUGHOUT

- (X, d) is a metric space, and $x, y, z \in X$ are points.
- $B(x, r) = \{y \in X : d(x, y) < r\}$ is the open ball and $\bar{B}(x, r) = \{y : d(x, y) \le r\}$ is the closed ball.
- $\operatorname{int} A$ or A° is the interior, $\operatorname{cl} A$ or $\overline{A}$ is the closure, ∂A is the boundary, and $A^c = X \setminus A$ is the complement.
- (x_n) is a sequence, and $x_n \to x$ means it converges to x.
- $\mathbb{N} = \{1, 2, 3, \ldots\}$, and $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are the integers, rationals, reals and complex numbers.

Chapter 0: Preliminaries

⏱ 2:10–2:30 (20 min)

Read quickly. These are tools you'll use in every later chapter. Make sure you can apply the preimage rules and negate quantifiers without thinking.

0.1 Logic and quantifiers

Almost every definition in this course is a chain of "for all" (∀) and "there exists" (∃). To **negate** a statement, swap ∀ ↔ ∃ and negate the final condition.

> **EXAMPLE: NEGATING CONVERGENCE**
>
> $x_n \to x$ means: $\forall \varepsilon > 0\ \exists N\ \forall n \geq N : d(x_n, x) < \varepsilon$.
>
> $x_n \not\to x$ means: $\exists \varepsilon > 0\ \forall N\ \exists n \geq N : d(x_n, x) \geq \varepsilon$. In words, infinitely many terms stay at least ε away from x.

Proof methods you will use: **direct**, **contrapositive** (to prove $P \Rightarrow Q$, prove not $Q \Rightarrow$ not P), **contradiction**, and **induction**.

0.2 Sets

- **De Morgan's laws:** $(\bigcup A_i)^c = \bigcap A_i^c$ and $(\bigcap A_i)^c = \bigcup A_i^c$. These are why "unions of open sets are open" turns into "intersections of closed sets are closed".
- $A \setminus B = A \cap B^c$.
- To prove $A = B$, show $A \subseteq B$ and $B \subseteq A$. Take an arbitrary element of one set and show it lies in the other.

0.3 Functions, images and preimages

DEFINITIONS

Let $f : X \to Y$.

- The **image** of $A \subseteq X$ is $f(A) = \{f(a) : a \in A\}$.
- The **preimage** (inverse image) of $B \subseteq Y$ is $f^{-1}(B) = \{x \in X : f(x) \in B\}$. This makes sense even when f has no inverse.
- f is **injective** (one-to-one) if $f(x) = f(y) \Rightarrow x = y$.
- f is **surjective** (onto) if every $y \in Y$ equals $f(x)$ for some x.
- f is **bijective** if it is both. Exactly then there is an inverse function $f^{-1} : Y \to X$.

PREIMAGES PRESERVE EVERYTHING (MEMORISE THIS)

- $f^{-1}(\bigcup B_i) = \bigcup f^{-1}(B_i)$
- $f^{-1}(\bigcap B_i) = \bigcap f^{-1}(B_i)$
- $f^{-1}(Y \setminus B) = X \setminus f^{-1}(B)$

Images are less well behaved: $f(\bigcup A_i) = \bigcup f(A_i)$, but only $f(A \cap B) \subseteq f(A) \cap f(B)$ (equality holds if f is injective). Also $A \subseteq f^{-1}(f(A))$ and $f(f^{-1}(B)) \subseteq B$.

WHY THIS MATTERS

Continuity is defined using **preimages** ("the preimage of an open set is open") because preimages respect complements, unions and intersections. Images don't, which is why "the image of an open set is open" is **not** the definition of continuity.

0.4 Countability

A set is **countable** if it is finite or in bijection with $\mathbb{N}$. $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$ are countable, and a countable union of countable sets is countable. $\mathbb{R}$ is **uncountable** (Cantor's diagonal argument). These facts come up with dense sets and separability.

0.5 The real numbers: supremum and infimum

DEFINITION

An **upper bound** of $A \subseteq \mathbb{R}$ is an M with $a \le M$ for all $a \in A$. The **supremum** $\sup A$ is the *least* upper bound. The **infimum** $\inf A$ is the greatest lower bound.

Completeness axiom of $\mathbb{R}$: every non-empty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$. This is exactly what $\mathbb{Q}$ lacks. For example, $\{q \in \mathbb{Q} : q^2 < 2\}$ has no supremum in $\mathbb{Q}$.

APPROXIMATION PROPERTY

If $s = \sup A$, then for every $\varepsilon > 0$ there is an $a \in A$ with $s - \varepsilon < a \le s$. You'll use this all the time, for example with $d(x, A) = \inf$ and with diameters.

0.6 Inequalities

- **Triangle inequality:** $|a + b| \le |a| + |b|$.
- **Reverse triangle inequality:** $||a| - |b|| \le |a - b|$.
- **Cauchy–Schwarz:** $(\Sigma\, a_i b_i)^2 \le (\Sigma\, a_i^2)(\Sigma\, b_i^2)$.
- **Minkowski:** $\sqrt{\Sigma(a_i+b_i)^2} \le \sqrt{\Sigma a_i^2} + \sqrt{\Sigma b_i^2}$. This is the triangle inequality for the Euclidean metric.

0.7 Sequences in $\mathbb{R}$ (you'll generalise all of this)

- $x_n \to x \Leftrightarrow \forall \varepsilon > 0\ \exists N\ \forall n \ge N : |x_n - x| < \varepsilon$.
- **Monotone convergence theorem:** a bounded monotone sequence of reals converges.
- **Bolzano–Weierstrass:** every bounded sequence in $\mathbb{R}$ (or $\mathbb{R}^n$) has a convergent subsequence.
- A **subsequence** (x_{n_k}) uses indices $n_1 < n_2 < n_3 < \ldots$, so $n_k \ge k$.

Chapter 1: Metric spaces

⏱ 2:30–3:05 (35 min)

Learn the three axioms by heart. Work through the standard examples, especially the $\mathbb{R}^n$ metrics, the discrete metric and C[a,b]. Practise one "prove this is a metric" proof and one "show this is NOT a metric" counterexample.

1.1 The big idea

Intuition. Analysis on $\mathbb{R}$ (limits, continuity, convergence) only ever uses one thing: the **distance** $|x - y|$. A metric space is any set with a sensible notion of distance. Once we have one, we can do analysis on sets of points, vectors, functions, strings, and so on.

DEFINITION 1.1: METRIC SPACE

A **metric** on a set X is a function $d : X \times X \to \mathbb{R}$ such that for all $x, y, z \in X$:

1. **(M1) Positivity:** $d(x, y) \geq 0$, and $d(x, y) = 0 \Leftrightarrow x = y$.
2. **(M2) Symmetry:** $d(x, y) = d(y, x)$.
3. **(M3) Triangle inequality:** $d(x, z) \leq d(x, y) + d(y, z)$.

The pair (X, d) is a **metric space**.

EXAM TIP

When asked to "prove d is a metric", check all three axioms explicitly and in order. M1 and M2 are usually one line each. All the work goes into M3.

1.2 Standard examples (know all of these)

(a) The real line

$X = \mathbb{R}$, $d(x, y) = |x - y|$. This is the **usual** or **standard** metric.

(b) Three metrics on $\mathbb{R}^n$

For $x = (x_1,\ldots,x_n)$ and $y = (y_1,\ldots,y_n)$:

$$d_1(x,y) = \Sigma_i |x_i - y_i| \text{ (taxicab / Manhattan)}$$

$$d_2(x,y) = \sqrt{\left(\Sigma_i\ (x_i - y_i)^2\right)}\ \text{(Euclidean)}$$

$$d_\infty(x,y) = \max_i |x_i - y_i|\ \text{(sup / max / Chebyshev)}$$

M1 and M2 are immediate. For M3: d_1 and d_∞ follow from the triangle inequality in $\mathbb{R}$, applied coordinate by coordinate. d_2 needs Minkowski / Cauchy–Schwarz.

> **COMPARISON INEQUALITIES IN $\mathbb{R}^n$**
>
> $$d_\infty(x,y) \le d_2(x,y) \le d_1(x,y) \le n \cdot d_\infty(x,y), \text{ and } d_2 \le \sqrt{n} \cdot d_\infty.$$
>
> These show the three metrics are **Lipschitz equivalent** (§1.5). So they have the same open sets, the same convergent sequences and the same continuous functions.

(c) The discrete metric

On **any** set X: $d(x, y) = 0$ if $x = y$ and $d(x, y) = 1$ if $x \neq y$.

> ***Proof.*** M1 and M2 are clear. For M3, if $x = z$ the left side is $0 \le$ anything. If $x \neq z$, then y can't equal both x and z, so at least one term on the right is 1. Then $d(x,z) = 1 \le d(x,y) + d(y,z)$. ∎

> **WHY THE DISCRETE METRIC MATTERS**
>
> It's the go-to **counterexample machine**. In it, every set is open and closed, every function out of it is continuous, convergent sequences are eventually constant, and it is always complete. It is compact only if X is finite. Whenever you suspect a statement is false in general, test it here first.

(d) Function spaces C[a, b]

C[a,b] is the set of continuous functions $f : [a,b] \to \mathbb{R}$.

$$d_\infty(f,g) = \sup_{t \in [a,b]} |f(t) - g(t)|\ \text{(uniform / sup metric)}$$

$$d_1(f,g) = \int_a^b |f(t) - g(t)|\, dt\ (L^1 \text{ / integral metric})$$

The sup is finite because $|f - g|$ is continuous on a closed bounded interval, so it attains a maximum. For d_1, **continuity is needed for M1**: if $f \neq g$ then $|f - g| > 0$ on a small interval (by continuity), so the integral is > 0. On merely integrable functions, d_1 fails M1. For example, a function that is 0 except at one point has integral 0.

Picture: $d_\infty(f,g)$ is the largest vertical gap between the graphs. $d_1(f,g)$ is the area between them.

(e) Sequence spaces

- ℓ^1: sequences with $\Sigma|x_i| < \infty$, with $d(x,y) = \Sigma|x_i - y_i|$.
- ℓ^2: sequences with $\Sigma x_i^2 < \infty$, with $d(x,y) = \sqrt{\Sigma(x_i - y_i)^2}$.
- ℓ^∞: bounded sequences, with $d(x,y) = \sup|x_i - y_i|$.

(f) Normed spaces

DEFINITION 1.2: NORM

A **norm** on a real vector space V is a map $\|\cdot\| : V \to \mathbb{R}$ such that:

1. $\|v\| \geq 0$, and $\|v\| = 0 \Leftrightarrow v = 0$;
2. $\|\lambda v\| = |\lambda| \|v\|$;
3. $\|u + v\| \leq \|u\| + \|v\|$.

Every norm gives a metric $d(u, v) = \|u - v\|$. All the examples in (b), (d) and (e) come from norms. The discrete metric does **not** come from a norm on any non-zero vector space, because $d(\lambda v, 0)$ would have to equal $|\lambda| \, d(v, 0)$.

(g) Other examples that come up in exams

- **Hamming metric:** on strings of length n, $d(x,y)$ is the number of positions where x and y differ.
- **Bounded metric:** $d'(x,y) = d(x,y) / (1 + d(x,y))$ is a metric, and every distance is < 1. It is proved in §1.6.
- **British Rail / SNCF metric** on $\mathbb{R}^2$: $d(x,y) = \|x - y\|$ if x, y and the origin are collinear, and $\|x\| + \|y\|$ otherwise. All journeys go via London (the origin).
- **Subspace (induced) metric:** if $Y \subseteq X$, restricting d to $Y \times Y$ makes Y a metric space. For example, $\mathbb{Q}$ or $[0,1]$ with $|x - y|$.
- **Product metric:** on $X \times Y$, use $\max\{d_X(x,x'), d_Y(y,y')\}$ or $d_X + d_Y$.

1.3 Things that are NOT metrics

STANDARD FAILURES

- $d(x,y) = (x - y)^2$ on $\mathbb{R}$ fails **M3**: $d(0,2) = 4 > d(0,1) + d(1,2) = 2$.
- $d(x,y) = |x^2 - y^2|$ on $\mathbb{R}$ fails **M1**: $d(1, -1) = 0$ but $1 \neq -1$.
- $d(x,y) = x - y$ fails **M1** (it can be negative) and **M2**.
- $d(f,g) = |f(0) - g(0)|$ on $C[0,1]$ fails **M1**: two different functions can agree at 0. (This is called a **pseudometric**.)

To disprove an axiom you only need **one specific counterexample with numbers**.

1.4 Useful consequences of the axioms

LEMMA 1.3

1. **Reverse triangle inequality:** $|d(x,z) - d(y,z)| \le d(x,y)$.
2. **Polygon inequality:** $d(x_1, x_n) \le d(x_1,x_2) + d(x_2,x_3) + \dots + d(x_{n-1}, x_n)$.
3. $|d(x,y) - d(x',y')| \le d(x,x') + d(y,y')$.

Proof. For (1): by M3, $d(x,z) \le d(x,y) + d(y,z)$, so $d(x,z) - d(y,z) \le d(x,y)$. Swapping x and y (and using M2) gives $d(y,z) - d(x,z) \le d(x,y)$. Together these give the absolute value. ∎

DEFINITION 1.4: BOUNDED, DIAMETER, DISTANCE TO A SET

- $A \subseteq X$ is **bounded** if there are $x_0 \in X$ and M with $d(a, x_0) \le M$ for all $a \in A$. Equivalently, A lies inside some ball.
- **Diameter:** $\operatorname{diam} A = \sup\{d(a,b) : a, b \in A\}$. A is bounded $\Leftrightarrow$ $\operatorname{diam} A < \infty$.
- **Distance from a point to a set:** $d(x, A) = \inf\{d(x,a) : a \in A\}$.
- **Distance between sets:** $d(A, B) = \inf\{d(a,b) : a \in A, b \in B\}$.

COMMON MISTAKES

- $d(x, A) = 0$ does **not** mean $x \in A$. For example, $d(0, (0,1)) = 0$. It means x is in the **closure** of A (Chapter 2).
- "Bounded" depends on the metric. $\mathbb{R}$ is unbounded under $|x-y|$ but bounded under the discrete metric or $d/(1+d)$.

DEFINITION 1.5: ISOMETRY

A map $f : (X,d_X) \to (Y,d_Y)$ is an **isometry** if $d_Y(f(x), f(y)) = d_X(x,y)$ for all x, y. Isometries are automatically injective, by M1.

1.5 Equivalent metrics

DEFINITION 1.6

- d and d' on X are **Lipschitz (strongly) equivalent** if there are constants $\alpha, \beta > 0$ with

$$\alpha \cdot d(x,y) \le d'(x,y) \le \beta \cdot d(x,y) \text{ for all } x, y.$$

- They are **topologically equivalent** if they give the same open sets (Chapter 2). Equivalently, the identity map is continuous in both directions.

FACT

Lipschitz equivalent ⇒ topologically equivalent, but **not conversely**. On $\mathbb{R}$, $d = |x-y|$ and $d' = d/(1+d)$ are topologically equivalent but not Lipschitz equivalent. If $d \le \beta\, d'$ held, d would be bounded, because $d' < 1$.

EXAMPLE: D_1 AND D_∞ ON C[0,1] ARE NOT EQUIVALENT

Always $d_1(f,g) \le d_\infty(f,g)$, because the area is at most (height × width 1). There is no reverse inequality. Take $f_n(t) = t^n$: then $d_1(f_n, 0) = 1/(n+1) \to 0$, but $d_\infty(f_n, 0) = 1$ for all n. So $f_n \to 0$ in d_1 but not in d_∞.

1.6 Worked proof: d/(1+d) is a metric

CLAIM

If d is a metric on X, then $\rho(x,y) = d(x,y) / (1 + d(x,y))$ is a metric on X.

Proof. Let $\varphi(t) = t/(1+t) = 1 - 1/(1+t)$ for $t \ge 0$. Then $\varphi(t) \ge 0$, $\varphi(t) = 0 \iff t = 0$, and φ is **increasing**. So M1 and M2 for ρ follow from those for d.

Subadditivity: for $a, b \ge 0$,

$$\varphi(a+b) = a/(1+a+b) + b/(1+a+b) \le a/(1+a) + b/(1+b) = \varphi(a) + \varphi(b).$$

M3: since $d(x,z) \le d(x,y) + d(y,z)$ and φ is increasing,

$$\rho(x,z) = \varphi(d(x,z)) \le \varphi(d(x,y) + d(y,z)) \le \varphi(d(x,y)) + \varphi(d(y,z)) = \rho(x,y) + \rho(y,z).$$

∎

GENERAL RECIPE

If $\varphi : [0,\infty) \to [0,\infty)$ is increasing, subadditive, and $\varphi(t) = 0 \iff t = 0$, then $\varphi \circ d$ is a metric. Examples are $\sqrt{d}$, $\min\{d, 1\}$ and $d/(1+d)$, but **not** d^2.

1.7 Practice questions

EASY 1. Show that $d(x,y) = |x^3 - y^3|$ is a metric on $\mathbb{R}$.

Solution

M1: $|x^3 - y^3| \geq 0$, and it is $0 \Leftrightarrow x^3 = y^3 \Leftrightarrow x = y$, since cubing is injective on $\mathbb{R}$. M2 is clear. M3: $|x^3 - z^3| = |(x^3 - y^3) + (y^3 - z^3)| \leq |x^3 - y^3| + |y^3 - z^3|$. In general, $|f(x) - f(y)|$ is a metric whenever f is injective. Compare $|x^2 - y^2|$, which fails because squaring isn't injective.

EASY 2. Compute d_1, d_2, d_∞ between $(1, 2)$ and $(4, -2)$ in $\mathbb{R}^2$.

Solution

The differences are 3 and 4. So $d_1 = 7$, $d_2 = 5$ and $d_\infty = 4$. Check: $4 \leq 5 \leq 7 \leq 2 \cdot 4$. ✓

MEDIUM 3. Is $d(x,y) = \min\{1, |x - y|\}$ a metric on $\mathbb{R}$? Is it Lipschitz equivalent to the usual metric?

Solution

Yes, it's a metric. $\varphi(t) = \min\{1,t\}$ is increasing, satisfies $\varphi(t) = 0 \Leftrightarrow t = 0$, and is subadditive: if a or $b \geq 1$ the right side is $\geq 1 \geq \varphi(a+b)$; otherwise $\varphi(a+b) \leq a + b = \varphi(a) + \varphi(b)$. It is **not** Lipschitz equivalent to $|x - y|$: we would need $|x-y| \leq \beta \cdot \min\{1,|x-y|\} \leq \beta$ for all x, y, which is impossible. It is topologically equivalent, because small balls (radius < 1) are the same in both metrics.

HARD 4. Let p be prime. For $x \in \mathbb{Z} \setminus \{0\}$, let $v(x)$ be the largest power k with $p^k \mid x$. Define $d(x,y) = p^{-v(x-y)}$ for $x \neq y$ and $d(x,x) = 0$. Show that d satisfies the **ultrametric inequality** $d(x,z) \leq \max\{d(x,y), d(y,z)\}$, and hence is a metric.

Solution

M1 and M2 are clear ($v(-a) = v(a)$). If p^k divides both $x - y$ and $y - z$, it divides their sum $x - z$. So $v(x - z) \geq \min\{v(x-y), v(y-z)\}$, and therefore $d(x,z) = p^{-v(x-z)} \leq \max\{d(x,y), d(y,z)\}$. (The cases where two of the points are equal are trivial.) Since $\max\{a,b\} \leq a + b$, M3 follows. A strange consequence: in an ultrametric space, every point of a ball is its centre.

HARD 5. Prove that d_2 satisfies the triangle inequality on $\mathbb{R}^n$.

Solution

Let $a = x - y$ and $b = y - z$, so $x - z = a + b$. We need $\|a+b\| \le \|a\| + \|b\|$, where $\|a\| = \sqrt{\Sigma a_i^2}$. Square both sides:

$$\|a+b\|^2 = \Sigma a_i^2 + 2\Sigma a_i b_i + \Sigma b_i^2 \le \|a\|^2 + 2\|a\|\|b\| + \|b\|^2 = (\|a\|+\|b\|)^2,$$

using Cauchy–Schwarz, $\Sigma a_i b_i \le \|a\|\|b\|$. Both sides are non-negative, so take square roots.

Chapter 2: Open and closed sets, convergence, topological spaces

⏱ 3:15–3:50 (35 min)

This is the most terminology-heavy chapter, and the language every later chapter uses. Learn the definitions of open, closed, interior, closure, boundary, limit point and dense word for word. Fill in the worked table in §2.4 yourself before checking it.

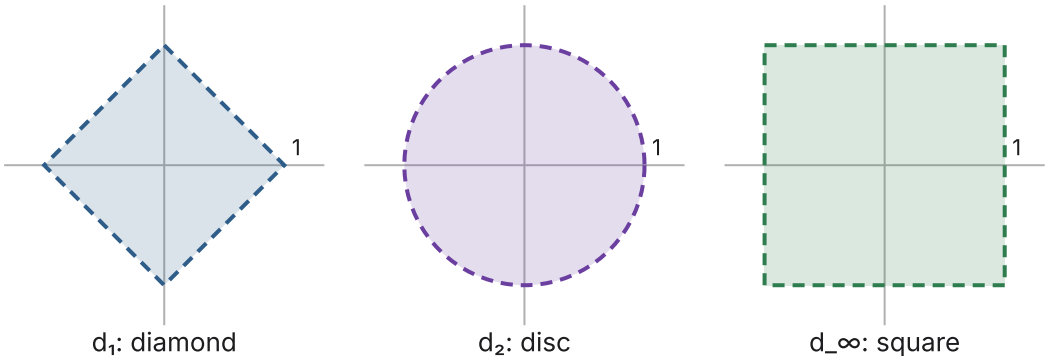
2.1 Balls

DEFINITION 2.1

For $x \in X$ and $r > 0$:

- **Open ball:** $B(x, r) = \{y \in X : d(x,y) < r\}$
- **Closed ball:** $\bar{B}(x, r) = \{y \in X : d(x,y) \le r\}$
- **Sphere:** $S(x, r) = \{y \in X : d(x,y) = r\}$

Balls depend on the metric. Here is the unit ball $B(0,1)$ in $\mathbb{R}^2$ for the three standard metrics:



Dashed edges show the boundary is *not* included in the open ball. Note that diamond $\subseteq$ disc $\subseteq$ square, which is the inequality $d_\infty \le d_2 \le d_1$ seen through balls.

EXAMPLE: BALLS IN THE DISCRETE METRIC

$B(x, r) = \{x\}$ if $r \le 1$, and $B(x, r) = X$ if $r > 1$. Also $\bar{B}(x, 1) = X$ while $B(x,1) = \{x\}$.

2.2 Open and closed sets

DEFINITION 2.2: OPEN SET

$U \subseteq X$ is **open** if for every $x \in U$ there is an $r > 0$ with $B(x, r) \subseteq U$.

Intuition: every point of U has some breathing room. You can move a little in any direction and stay in U. No point of U sits on its edge.

DEFINITION 2.3: CLOSED SET

$F \subseteq X$ is **closed** if its complement $X \setminus F$ is open.

Equivalently (Theorem 2.10), F contains the limit of every convergent sequence of points in F: you can't escape F by taking limits.

LEMMA 2.4: OPEN BALLS ARE OPEN

For any x and $r > 0$, $B(x, r)$ is open. Closed balls and spheres are closed.

Proof. Let $y \in B(x,r)$ and put $s = r - d(x,y) > 0$. If $z \in B(y, s)$, then $d(x,z) \le d(x,y) + d(y,z) < d(x,y) + s = r$, so $z \in B(x,r)$. Hence $B(y,s) \subseteq B(x,r)$.
For closed balls, show the complement is open. If $d(x,y) > r$, put $s = d(x,y) - r$. For $z \in B(y,s)$, $d(x,z) \ge d(x,y) - d(y,z) > d(x,y) - s = r$. ∎

EXAM TIP: THE "R – D(X,Y)" TRICK

This radius choice is the single most reused idea in the course. Draw the picture: a small ball around y that fits inside the big ball.

THEOREM 2.5: PROPERTIES OF OPEN SETS

1. $\varnothing$ and X are open.
2. Any union (even infinitely many) of open sets is open.
3. A **finite** intersection of open sets is open.

Taking complements (De Morgan): $\varnothing$, X are closed, **any** intersection of closed sets is closed, and a **finite** union of closed sets is closed.

Proof. (2): if $x \in \bigcup U_i$, then $x \in U_j$ for some j, and some $B(x,r) \subseteq U_j \subseteq \bigcup U_i$. (3): if $x \in U_1 \cap \dots \cap U_n$, choose $B(x, r_i) \subseteq U_i$ and let $r = \min\{r_1,\dots,r_n\} > 0$. The minimum of **finitely many** positive numbers is positive. This step

fails for infinitely many: an infimum can be 0. ∎

WHY "FINITE" MATTERS: COUNTEREXAMPLES

- $\bigcap_n (-1/n, 1/n) = \{0\}$: infinitely many open sets whose intersection is not open in $\mathbb{R}$.
- $\bigcup_n [1/n, 1] = (0, 1]$: infinitely many closed sets whose union is not closed.

THE MOST COMMON MISCONCEPTION: SETS ARE NOT DOORS

"Not open" does **not** mean "closed". A set can be:

- **both** open and closed ("clopen"): $\varnothing$ and X always, and every subset in a discrete space;
- **neither**: $[0,1)$ in $\mathbb{R}$, or $\mathbb{Q}$ in $\mathbb{R}$;
- exactly one of the two.

EXAMPLES IN $\mathbb{R}$ (USUAL METRIC)

Set	Open?	Closed?	Reason
(a, b)	✓	×	It's a ball. Points $x_n \to a$ inside it have a limit outside it.
$[a, b]$	×	✓	Every ball around a leaves the set. The complement is a union of open intervals.
$[a, b)$	×	×	Fails at a for openness, and at b for closedness.
$\{p\}$	×	✓	Finite sets are always closed in a metric space.
$\mathbb{Z}$	×	✓	The complement is a union of intervals $(n, n+1)$.
$\mathbb{Q}$	×	×	Every interval contains both rationals and irrationals.
$\{1/n : n \in \mathbb{N}\}$	×	×	$1/n \to 0$, which is not in the set. Adding $\{0\}$ makes it closed.
$\mathbb{R}, \varnothing$	✓	✓	These are always clopen.

OPEN DEPENDS ON THE AMBIENT SPACE

$[0, 1)$ is not open in $\mathbb{R}$, but it **is** open in the subspace $X = [0, 2]$, because $[0,1) = B_X(0, 1)$. Always ask: open *in what*? (See §2.7.)

2.3 Interior, closure, boundary, limit points

DEFINITION 2.6

Let $A \subseteq X$.

- **Interior point** of A: an x with $B(x,r) \subseteq A$ for some $r > 0$.
  Interior int A: the set of all interior points. It is the **largest open set contained in** A, which equals the union of all open subsets of A.
- **Closure** cl A: the **smallest closed set containing** A, which equals the intersection of all closed supersets of A.
- **Limit point** (accumulation point, cluster point) of A: an x such that every ball B(x,r) contains a point of A **other than x**. The set of limit points is the **derived set** A'.
- **Isolated point** of A: a point $a \in A$ with $B(a,r) \cap A = \{a\}$ for some r.
- **Boundary**: $\partial A = \text{cl } A \setminus \text{int } A$.
- **Exterior**: $\text{ext } A = \text{int}(X \setminus A)$.

THEOREM 2.7: CHARACTERISATIONS OF CLOSURE (ALL EQUIVALENT; VERY EXAMINABLE)

For $x \in X$, the following are equivalent:

1. $x \in \text{cl } A$;
2. every open ball B(x, r) meets A (that is, $B(x,r) \cap A \neq \varnothing$);
3. there is a sequence (a_n) in A with $a_n \to x$;
4. $d(x, A) = 0$;
5. $x \in A \cup A'$.

Also: A is open $\Leftrightarrow$ $A = \text{int } A$; A is closed $\Leftrightarrow$ $A = \text{cl } A \Leftrightarrow A' \subseteq A$.

Proof. (1)⇒(2): if some B(x,r) misses A, then $X \setminus B(x,r)$ is a closed set containing A, so it contains cl A. That gives $x \notin \text{cl } A$. (2)⇒(3): pick $a_n \in B(x, 1/n) \cap A$. Then $d(a_n, x) < 1/n \to 0$. (3)⇒(4): $d(x,A) \le d(x,a_n) \to 0$. (4)⇒(2): for any r, by the definition of inf there is a $\in A$ with $d(x,a) < r$. (2)⇒(1): let E be the set of points satisfying (2). Its complement is open: if $B(y,r) \cap A = \varnothing$, then every $z \in B(y,r)$ has a small ball inside B(y,r), which misses A. So E is a closed set containing A, and hence $\text{cl } A \subseteq E$. ∎

THEOREM 2.8: USEFUL IDENTITIES

- $X \setminus \operatorname{cl} A = \operatorname{int}(X \setminus A)$ and $X \setminus \operatorname{int} A = \operatorname{cl}(X \setminus A)$.
- $\partial A = \operatorname{cl} A \cap \operatorname{cl}(X \setminus A)$. In words, $x \in \partial A \Leftrightarrow$ every ball around x meets both A and its complement. ∂A is always closed, and $\partial A = \partial(X \setminus A)$.
- $\operatorname{cl}(A \cup B) = \operatorname{cl} A \cup \operatorname{cl} B$, but only $\operatorname{cl}(A \cap B) \subseteq \operatorname{cl} A \cap \operatorname{cl} B$. For example, with $A = \mathbb{Q}$ and $B = \mathbb{R} \setminus \mathbb{Q}$, the left side is $\varnothing$ and the right side is $\mathbb{R}$.
- $\operatorname{int}(A \cap B) = \operatorname{int} A \cap \operatorname{int} B$, but only $\operatorname{int}(A \cup B) \supseteq \operatorname{int} A \cup \operatorname{int} B$.
- $A \subseteq B \Rightarrow \operatorname{int} A \subseteq \operatorname{int} B$ and $\operatorname{cl} A \subseteq \operatorname{cl} B$.

TRAP: CLOSURE OF AN OPEN BALL ≠ CLOSED BALL (IN GENERAL)

In the discrete metric, $\operatorname{cl} B(x,1) = \operatorname{cl}\{x\} = \{x\}$, but $\bar{B}(x,1) = X$. What is always true is $\operatorname{cl} B(x,r) \subseteq \bar{B}(x,r)$. Equality holds in normed spaces such as $\mathbb{R}^n$ and $C[a,b]$.

DEFINITION 2.9: DENSE, SEPARABLE, NOWHERE DENSE

- A is **dense** in X if $\operatorname{cl} A = X$. Equivalently, every non-empty open set meets A, or every point is a limit of points of A. Examples: $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$; polynomials are dense in $(C[a,b], d_\infty)$ (Weierstrass).
- X is **separable** if it has a **countable** dense subset. Examples: $\mathbb{R}^n$ (via $\mathbb{Q}^n$) and $C[a,b]$. An uncountable discrete space is **not** separable.
- A is **nowhere dense** if $\operatorname{int}(\operatorname{cl} A) = \varnothing$. Examples: $\mathbb{Z}$ or the Cantor set in $\mathbb{R}$.

2.4 Worked example: compute everything

$A = [0, 1) \cup \{2\} \cup (\mathbb{Q} \cap (3, 4))$ IN $\mathbb{R}$

Object	Answer	Why
int A	$(0, 1)$	Points of (0,1) have room around them. Balls around 0 or 2 leave A. Every interval around a rational contains irrationals.
cl A	$[0, 1] \cup \{2\} \cup [3, 4]$	1 is a limit of points in A. The rationals are dense in [3,4].
∂A	$\{0, 1, 2\} \cup [3, 4]$	cl A minus int A
A' (limit points)	$[0, 1] \cup [3, 4]$	2 is not a limit point because it is isolated.
Isolated points	$\{2\}$	$B(2, ½) \cap A = \{2\}$
Open? Closed?	Neither	$0 \in A$ is not an interior point, and $1 \in \text{cl } A \setminus A$.

2.5 Convergence of sequences

DEFINITION 2.10

$x_n \to x$ in (X, d) if $d(x_n, x) \to 0$ in $\mathbb{R}$. Explicitly:

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall n \geq N : d(x_n, x) < \varepsilon.$$

Equivalently, every open set (or ball) containing x contains x_n for all but finitely many n (the sequence is **eventually** in it).

THEOREM 2.11: BASIC FACTS

1. **Limits are unique** in a metric space.
2. A convergent sequence is bounded.
3. If $x_n \to x$, every subsequence also converges to x.
4. **Sequential closedness:** F is closed $\Leftrightarrow$ whenever $x_n \in F$ and $x_n \to x \in X$, we have $x \in F$.

Proof. (1): if $x_n \to x$ and $x_n \to y$, then $0 \le d(x,y) \le d(x,x_n) + d(x_n,y) \to 0$, so $d(x,y) = 0$ and $x = y$. (4) $\Rightarrow$: suppose $x \notin F$. Since $X \setminus F$ is open, some $B(x,r)$ misses F, so $d(x_n, x) \ge r$ for all n. That contradicts $x_n \to x$. (4) $\Leftarrow$: by Theorem 2.7(3), any $x \in \operatorname{cl} F$ is a limit of points of F, so $x \in F$. Hence $\operatorname{cl} F = F$. ∎

WHAT CONVERGENCE MEANS IN SPECIFIC SPACES

- $\mathbb{R}^n$ (any of d_1, d_2, d_∞): convergence $\Leftrightarrow$ **each coordinate** converges.
- **$(C[a,b], d_\infty)$**: convergence $\Leftrightarrow$ **uniform convergence**.
- **$(C[a,b], d_1)$**: convergence "in mean". This is weaker: $t^n \to 0$ in d_1 but not in d_∞.
- **Discrete metric**: $x_n \to x \Leftrightarrow (x_n)$ is **eventually constant** and equal to x (take $\varepsilon = \tfrac{1}{2}$).

2.6 Topological spaces

Intuition. Notice that convergence, continuity and closure can all be described using only **which sets are open**. So we can forget the distances and just keep the collection of open sets. That collection is a **topology**.

DEFINITION 2.12: TOPOLOGY

A **topology** on a set X is a collection τ of subsets of X (called **open sets**) such that:

1. **(T1)** $\varnothing \in \tau$ and $X \in \tau$;
2. **(T2)** any union of members of τ is in τ;
3. **(T3)** any finite intersection of members of τ is in τ (it's enough to check intersections of two).

(X, τ) is a **topological space**. A set is **closed** if its complement is in τ. A **neighbourhood** of x is a set containing an open set that contains x.

STANDARD TOPOLOGIES

- **Metric topology**: the open sets of a metric space (Theorem 2.5 checks T1–T3).
- **Discrete topology**: τ is **every** subset of X. It comes from the discrete metric.
- **Indiscrete (trivial) topology**: $\tau = \{\varnothing, X\}$.
- **Cofinite topology**: U is open $\Leftrightarrow U = \varnothing$ or $X \setminus U$ is finite.
- **Sierpiński space**: $X = \{0,1\}$, $\tau = \{\varnothing, \{1\}, \{0,1\}\}$.

EXAMPLE: CHECKING A TOPOLOGY

Is $\tau = \{\varnothing, \{a\}, \{b\}, \{a,b,c\}\}$ a topology on $\{a,b,c\}$? **No**: $\{a\} \cup \{b\} = \{a,b\} \notin \tau$, so T2 fails. Adding $\{a,b\}$ fixes it.

DEFINITION 2.13: COMPARING TOPOLOGIES, BASIS, METRIZABLE, HAUSDORFF

- τ_1 is **coarser** (weaker) than τ_2 if $\tau_1 \subseteq \tau_2$; then τ_2 is **finer** (stronger). The indiscrete topology is the coarsest possible and the discrete topology is the finest.
- A **basis** $\mathcal{B}$ for τ is a collection of open sets such that every open set is a union of members of $\mathcal{B}$. Example: open balls are a basis for a metric topology, and intervals with rational endpoints are a basis for $\mathbb{R}$.
- (X,τ) is **metrizable** if some metric on X produces exactly τ.
- (X,τ) is **Hausdorff** (T_2) if any two distinct points $x \neq y$ have **disjoint** open neighbourhoods $U \ni x$ and $V \ni y$.

THEOREM 2.14: EVERY METRIC SPACE IS HAUSDORFF

Proof. If $x \neq y$, let $r = d(x,y)/2 > 0$. If z were in $B(x,r) \cap B(y,r)$, then $d(x,y) \leq d(x,z) + d(z,y) < 2r = d(x,y)$, a contradiction. ∎

EXAM USE: PROVING A SPACE IS NOT METRIZABLE

Show it isn't Hausdorff. The indiscrete topology on ≥ 2 points, the cofinite topology on an infinite set (any two non-empty open sets intersect) and Sierpiński space are all non-Hausdorff, so none of them is metrizable.

WEIRD BEHAVIOUR IN NON-HAUSDORFF SPACES

In the indiscrete topology, **every** sequence converges to **every** point, so limits aren't unique. Uniqueness of limits really comes from the Hausdorff property. Also, in general topological spaces, sequences aren't always enough to detect closure. That's why topology uses open sets as the basic notion.

2.7 Subspaces

DEFINITION 2.15: SUBSPACE TOPOLOGY

If $Y \subseteq X$, the **subspace topology** on Y is $\tau_Y = \{U \cap Y : U \text{ open in } X\}$. For metric spaces this is the same as the topology of the restricted metric, and $B_Y(y,r) = B_X(y,r) \cap Y$.

FACTS

- $F \subseteq Y$ is closed in $Y \Leftrightarrow F = C \cap Y$ for some C closed in X.
- If Y is open in X, then "open in Y" $\Leftrightarrow$ "open in X" for subsets of Y. The same holds for closed sets when Y is closed.
- Example: in $Y = \mathbb{Q}$, the set $\{q \in \mathbb{Q} : q^2 < 2\}$ is **clopen**. It equals $(-\sqrt{2}, \sqrt{2}) \cap \mathbb{Q} = [-\sqrt{2}, \sqrt{2}] \cap \mathbb{Q}$.

2.8 Practice questions

EASY 1. Prove that $U = \{(x,y) \in \mathbb{R}^2 : x > 0\}$ is open, directly from the definition.

Solution

Let $p = (a,b) \in U$, so $a > 0$. Take $r = a$. If $(x,y) \in B(p, a)$ (Euclidean), then $|x - a| \leq d_2((x,y),(a,b)) < a$, so $x > 0$. Hence $B(p,a) \subseteq U$. (Quicker method in Ch 3: $U = f^{-1}((0,\infty))$ for the continuous map $f(x,y) = x$.)

EASY 2. Prove that every finite subset of a metric space is closed.

Solution

First, $\{p\}$ is closed. If $x \neq p$, then $B(x, d(x,p))$ doesn't contain p, so the complement of $\{p\}$ is open. A finite set is a finite union of singletons, and finite unions of closed sets are closed.

MEDIUM 3. In a discrete metric space, find int A, cl A and ∂A for any A.

Solution

Every set is open ($B(x,½) = \{x\} \subseteq A$) and so every set is also closed. Hence int $A = A$, cl $A = A$ and $\partial A = \varnothing$.

MEDIUM 4. Show that $\partial A = \varnothing \Leftrightarrow A$ is clopen.

Solution

$\partial A = \text{cl } A \setminus \text{int } A$ is empty $\Leftrightarrow \text{cl } A \subseteq \text{int } A$. Since $\text{int } A \subseteq A \subseteq \text{cl } A$ always, that holds $\Leftrightarrow \text{int } A = A = \text{cl } A \Leftrightarrow A$ is both open and closed.

HARD 5. Let X be infinite with the cofinite topology. (a) Show it's a topology. (b) Show it's not Hausdorff. (c) Which sequences converge, and to what?

Solution

(a) T1: $\varnothing$ by definition, and $X \setminus X = \varnothing$ is finite. T2: if the U_i are non-empty open sets, $X \setminus \bigcup U_i = \bigcap (X \setminus U_i)$ is contained in a finite set. T3: $X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$ is a union of two finite sets. (b) If U, V are non-empty open and disjoint, then $X = X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$ is finite, a contradiction. (c) Suppose no value is repeated infinitely often (for example, all x_n are distinct). Then $x_n \to x$ for **every** $x \in X$. An open $U \ni x$ misses only finitely many points, so only finitely many x_n are outside U. This is another example of non-unique limits.

HARD 6. Prove that in any metric space, $\operatorname{cl} B(x,r) \subseteq \bar{B}(x,r)$, and give an example where the inclusion is strict.

Solution

$\bar{B}(x,r)$ is closed (Lemma 2.4) and contains $B(x,r)$. The closure is the *smallest* such closed set, so the inclusion follows. For a strict example, use the discrete metric with $r = 1$ and $|X| \geq 2$: $\operatorname{cl} B(x,1) = \{x\} \neq X = \bar{B}(x,1)$. Another example: $X = [0,1] \cup [2,3] \subset \mathbb{R}$ with $x = 0$, $r = 2$. Then $\operatorname{cl} B = [0,1]$, but $\bar{B} = [0,1] \cup \{2\}$.

Chapter 3: Continuity and homeomorphisms

⏱ 3:50–4:25 (35 min)

Learn the four equivalent definitions of continuity and how to switch between them. Memorise the uniform continuity hierarchy and its counterexamples. Be able to show two spaces are or are not homeomorphic.

3.1 Four equivalent definitions

Intuition. f is continuous if "close inputs give close outputs". If you want f(y) within ε of f(x), you can guarantee it by taking y within δ of x.

DEFINITION 3.1: ε–Δ CONTINUITY

f : (X,d_X) → (Y,d_Y) is **continuous at a ∈ X** if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in X : d_X(x,a) < \delta \Rightarrow d_Y(f(x), f(a)) < \varepsilon.$$

In terms of balls: f(B(a,δ)) ⊆ B(f(a), ε). f is **continuous** if it is continuous at every point.

THEOREM 3.2: EQUIVALENT FORMS (CORE EXAM THEOREM)

For f : X → Y between metric spaces, the following are equivalent:

1. **(ε–δ)** f is continuous at every point.
2. **(Sequential)** $x_n \to x$ in X ⇒ $f(x_n) \to f(x)$ in Y.
3. **(Open sets)** V open in Y ⇒ $f^{-1}(V)$ open in X.
4. **(Closed sets)** C closed in Y ⇒ $f^{-1}(C)$ closed in X.

For topological spaces, (3) is the **definition** of continuity.

Proof. **(1)⇒(3):** let V be open and a ∈ $f^{-1}(V)$. Since f(a) ∈ V, choose ε with B(f(a),ε) ⊆ V, then δ from (1). Then f(B(a,δ)) ⊆ B(f(a),ε) ⊆ V, so B(a,δ) ⊆ $f^{-1}(V)$.
(3)⇒(1): B(f(a),ε) is open, so $f^{-1}(B(f(a),\varepsilon))$ is open and contains a. Hence it contains some B(a,δ), which is exactly the ε–δ statement.
(3) ⇔ (4): take complements, using $f^{-1}(Y \setminus C) = X \setminus f^{-1}(C)$.
(1)⇒(2): given ε, take δ from (1), then N with $d(x_n,x) < \delta$ for n ≥ N. Then $d(f(x_n),f(x)) < \varepsilon$.

(2)⇒(1), by contrapositive: if (1) fails at a, there is an ε such that for each $\delta = 1/n$ some x_n has $d(x_n,a) < 1/n$ but $d(f(x_n),f(a)) \geq \varepsilon$. Then $x_n \to a$ but $f(x_n) \not\to f(a)$. ∎

WHICH FORM SHOULD YOU USE?

- To show f **is** continuous with a formula: use ε–δ, or combine known continuous functions.
- To show f is **not** continuous: find **one sequence** $x_n \to x$ with $f(x_n) \not\to f(x)$, or one open set whose preimage isn't open.
- To show a **set is open or closed**: write it as f^{-1}(open) or f^{-1}(closed) for a continuous f. This is often a one-line proof.
- For abstract or topological questions: use the open-set form.

3.2 Building continuous functions

THEOREM 3.3

- Constant maps and identity maps $(X,d) \to (X,d)$ are continuous.
- Compositions of continuous maps are continuous: $(g \circ f)^{-1}(V) = f^{-1}(g^{-1}(V))$.
- If $f, g : X \to \mathbb{R}$ are continuous, so are $f + g$, fg, $|f|$, $\max(f,g)$, $\min(f,g)$, and f/g where $g \neq 0$.
- $f = (f_1,\ldots,f_n) : X \to \mathbb{R}^n$ is continuous $\Leftrightarrow$ each component f_i is continuous.
- **Any** function out of a discrete space is continuous, because every preimage is open.
- For fixed a, $x \mapsto d(x,a)$ is continuous. More generally, $x \mapsto d(x,A)$ is 1-Lipschitz: $|d(x,A) - d(y,A)| \leq d(x,y)$.
- Restrictions of continuous maps to subspaces are continuous.

ONE-LINE OPEN/CLOSED PROOFS

- $\{(x,y) : x^2 + y^2 < 1\}$ is open: it is $f^{-1}((-\infty,1))$ with $f(x,y) = x^2+y^2$ continuous.
- $\{(x,y) : xy = 1\}$ is closed: it is $g^{-1}(\{1\})$ with $g(x,y) = xy$, and $\{1\}$ is closed.
- $\{f \in C[0,1] : f(\tfrac{1}{2}) > 0\}$ is open in d_∞: the evaluation map $f \mapsto f(\tfrac{1}{2})$ is 1-Lipschitz, since $|f(\tfrac{1}{2}) - g(\tfrac{1}{2})| \leq d_\infty(f,g)$.
- The zero set $\{x : f(x) = 0\}$ of any continuous real-valued f is closed.

CONTINUITY DOES NOT PRESERVE OPEN OR CLOSED SETS UNDER IMAGES

- Open sets can map to non-open sets: $f(x) = x^2$ sends $(-1,1)$ to $[0,1)$.
- Closed sets can map to non-closed sets: arctan sends the closed set $\mathbb{R}$ to $(-\pi/2, \pi/2)$. Projecting the closed hyperbola $xy = 1$ to the x-axis gives $\mathbb{R} \setminus \{0\}$.

Only **preimages** behave well. (Exception: continuous images of **compact** sets are compact, see Ch 5.)

CONTINUITY DEPENDS ON THE METRICS

The identity $(\mathbb{R}, \text{discrete}) \to (\mathbb{R}, \text{usual})$ is continuous, because everything is open in the domain. The identity $(\mathbb{R}, \text{usual}) \to (\mathbb{R}, \text{discrete})$ is **not**: the preimage of the open set $\{0\}$ is $\{0\}$, which is not open in the usual topology. Likewise, the identity $(C[0,1], d_\infty) \to (C[0,1], d_1)$ is continuous (since $d_1 \le d_\infty$), but the reverse is not ($t^n \to 0$ in d_1 only).

3.3 Uniform continuity and Lipschitz maps

DEFINITION 3.4

- f is **uniformly continuous** if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in X : d(x,y) < \delta \Rightarrow d(f(x),f(y)) < \varepsilon.$$

  The key difference from continuity is that **δ depends only on ε, not on the point**. Look at where "$\forall x$" sits in the quantifiers.
- f is **Lipschitz** with constant L if $d(f(x),f(y)) \le L \cdot d(x,y)$ for all x, y. If $L < 1$, f is a **contraction** (Ch 4).

HIERARCHY

$$\text{Lipschitz} \Rightarrow \text{uniformly continuous} \Rightarrow \text{continuous}$$

Lipschitz ⇒ UC: take $\delta = \varepsilon/L$. Neither arrow reverses:

Function	Continuous	Uniformly cont.	Lipschitz
x^2 on $\mathbb{R}$	✓	×	×
$1/x$ on $(0,1)$	✓	×	×
$\sin(1/x)$ on $(0,1)$	✓	×	×
$\sqrt{x}$ on $[0,1]$	✓	✓ (compact domain)	× (slope unbounded near 0)
$\sin x$ on $\mathbb{R}$	✓	✓	✓ ($L = 1$)
x^2 on $[0,1]$	✓	✓	✓ ($L = 2$)

PROOF: X^2 IS NOT UNIFORMLY CONTINUOUS ON $\mathbb{R}$

Take $\varepsilon = 1$. For any $\delta > 0$, let $x = 1/\delta$ and $y = 1/\delta + \delta/2$. Then $|x - y| < \delta$, but $|x^2 - y^2| = |x - y||x + y| = (\delta/2)(2/\delta + \delta/2) > 1$.

Sequence version (often easier): f is **not** UC $\Leftrightarrow$ there exist $\varepsilon > 0$ and sequences x_n, y_n with $d(x_n,y_n) \to 0$ but $d(f(x_n),f(y_n)) \geq \varepsilon$. For x^2, take $x_n = n$ and $y_n = n + 1/n$: then $y_n^2 - x_n^2 = 2 + 1/n^2 \geq 2$.

WHY UNIFORM CONTINUITY MATTERS

A uniformly continuous map sends **Cauchy sequences to Cauchy sequences**. A merely continuous map need not: $1/x$ on $(0,1)$ sends the Cauchy sequence $1/n$ to n. Also, any continuous function on a **compact** metric space is uniformly continuous (Heine–Cantor, Ch 5).

3.4 Homeomorphisms

DEFINITION 3.5

A **homeomorphism** $f : X \to Y$ is a **bijection** such that f **and** f^{-1} are both continuous. Then X and Y are **homeomorphic**, written $X \cong Y$. A **topological property** is one preserved by homeomorphisms.

Intuition: a homeomorphism stretches and bends without tearing or gluing. Topologists treat a coffee mug and a doughnut as the same.

EXAMPLES

- $(-1, 1) \cong \mathbb{R}$ via $x \mapsto x / (1 - |x|)$, or $x \mapsto \tan(\pi x/2)$.
- Any two open intervals are homeomorphic (via a linear map, then the example above). Similarly any two closed intervals.
- Any isometry onto its image is a homeomorphism onto its image.
- If d, d' are topologically equivalent, the identity $(X,d) \to (X,d')$ is a homeomorphism.

A CONTINUOUS BIJECTION IS NOT AUTOMATICALLY A HOMEOMORPHISM

- $f : [0, 2\pi) \to S^1$, $t \mapsto (\cos t, \sin t)$ is a continuous bijection onto the circle. Its inverse is not continuous at $(1,0)$: points just below $(1,0)$ on the circle map to numbers near 2π, not near 0.
- The identity $(\mathbb{R}$, discrete$) \to (\mathbb{R}$, usual$)$.

It **is** automatic when the domain is compact and the codomain is Hausdorff (Ch 5).

TOPOLOGICAL VS NON-TOPOLOGICAL PROPERTIES

Topological (preserved by $\cong$)	NOT topological (depends on the metric)
open/closed sets, convergence of sequences, continuity, Hausdorff, compactness, connectedness, separability, number of isolated points	**boundedness** ($\mathbb{R} \cong (-1,1)$), **completeness** ($\mathbb{R} \cong (0,1)$, but only $\mathbb{R}$ is complete), Cauchy sequences, uniform continuity, diameter

PROVING $X \ncong Y$

Find a topological property one space has and the other lacks. For example, $[0,1] \ncong (0,1)$ because one is compact and the other isn't. $\mathbb{R} \ncong \mathbb{R}^2$ because removing one point disconnects $\mathbb{R}$ but not $\mathbb{R}^2$. $[0,1) \ncong (0,1)$ because removing the point 0 leaves $[0,1)$ connected, while removing any point disconnects $(0,1)$.

3.5 Practice questions

EASY 1. Prove from ε–δ that $f(x) = 3x + 2$ is continuous on $\mathbb{R}$.

Solution

Given $\varepsilon > 0$, let $\delta = \varepsilon/3$. If $|x - a| < \delta$, then $|f(x) - f(a)| = 3|x - a| < \varepsilon$. It's in fact 3-Lipschitz, so it's uniformly continuous.

EASY 2. Show that $f(x) = 1$ for $x \geq 0$ and $f(x) = 0$ for $x < 0$ is not continuous on $\mathbb{R}$, using two different methods.

Solution

Sequences: $x_n = -1/n \to 0$, but $f(x_n) = 0 \not\to 1 = f(0)$. **Open sets:** $f^{-1}((\tfrac{1}{2}, 2)) = [0, \infty)$, which is not open.

MEDIUM 3. Let $A \subseteq X$ be non-empty. Show that $|d(x,A) - d(y,A)| \leq d(x,y)$, and deduce that $\operatorname{cl} A = \{x : d(x,A) = 0\}$ is closed.

Solution

For any $a \in A$, $d(x,A) \leq d(x,a) \leq d(x,y) + d(y,a)$. Taking inf over a gives $d(x,A) \leq d(x,y) + d(y,A)$. Swapping x and y gives the absolute value. So $x \mapsto d(x,A)$ is 1-Lipschitz and hence continuous, and $\{x : d(x,A) = 0\}$ is the preimage of the closed set $\{0\}$.

MEDIUM 4. Show that $I : C[0,1] \to \mathbb{R}$, $I(f) = \int_0^1 f$, is continuous with respect to d_∞, and also with respect to d_1.

Solution

$|I(f) - I(g)| = |\int(f - g)| \leq \int|f - g| = d_1(f,g) \leq d_\infty(f,g)$. So I is 1-Lipschitz for both metrics, and therefore uniformly continuous.

HARD 5. Is the evaluation map $E(f) = f(0)$ continuous from $(C[0,1], d_1)$ to $\mathbb{R}$?

Solution

No. Let $f_n(t) = \max\{0, 1 - nt\}$ (a spike of height 1 at 0). Then $d_1(f_n, 0) = 1/(2n) \to 0$, so $f_n \to 0$. But $E(f_n) = 1 \not\to 0 = E(0)$. (With d_∞, it is continuous: $|f(0) - g(0)| \leq d_\infty(f,g)$.)

HARD 6. Let $f, g : X \to Y$ be continuous, with Y a metric space. Show that $\{x : f(x) = g(x)\}$ is closed. Deduce that if $f = g$ on a dense subset, then $f = g$ everywhere.

Solution

$h(x) = d_Y(f(x), g(x))$ is continuous, because it's a composition of continuous maps (or argue with sequences). The set is $h^{-1}(\{0\})$, which is closed. If it contains a dense set D, then it contains $\operatorname{cl} D = X$. Application: a continuous function on $\mathbb{R}$ is determined by its values on $\mathbb{Q}$.

Chapter 4: Completeness

⏱ 4:40–5:10 (30 min)

Learn the Cauchy definition, the list of complete and incomplete spaces, the "closed ⇔ complete" theorem, and the full proof of the Contraction Mapping Theorem. That proof is a favourite long exam question.

4.1 Cauchy sequences

Intuition. A sequence is Cauchy if its terms eventually get as close to **each other** as you like, so they look as if they're converging. Completeness says that whenever this happens, there really is a limit in the space. $\mathbb{Q}$ is incomplete because 1, 1.4, 1.41, 1.414, … "wants" to converge to $\sqrt{2}$, which isn't in $\mathbb{Q}$. The space has holes.

DEFINITION 4.1: CAUCHY SEQUENCE

(x_n) is **Cauchy** if

$$\forall \varepsilon > 0 \ \exists N \ \forall m, n \geq N : d(x_m, x_n) < \varepsilon.$$

Notice that, unlike the definition of convergence, this one **doesn't mention a limit**.

THEOREM 4.2: BASIC FACTS (WITH PROOFS, ALL SHORT)

1. Every convergent sequence is Cauchy.
2. Every Cauchy sequence is bounded.
3. A Cauchy sequence with a convergent subsequence converges, to the same limit.

Proof. (1): if $x_n \to x$, choose N with $d(x_n,x) < \varepsilon/2$ for $n \geq N$. Then $d(x_m,x_n) \leq d(x_m,x) + d(x,x_n) < \varepsilon$. (2): with $\varepsilon = 1$, all x_n with $n \geq N$ lie in B(x_N, 1). Only finitely many terms remain outside, so take the radius $1 + \max\{d(x_i, x_N) : i < N\}$. (3): let $x_{n_k} \to x$. Given ε, choose N from the Cauchy condition with $\varepsilon/2$, and k with $n_k \geq N$ and $d(x_{n_k}, x) < \varepsilon/2$. For $n \geq N$, $d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \varepsilon$. ∎

4.2 Complete metric spaces

DEFINITION 4.3

(X, d) is **complete** if every Cauchy sequence in X converges to a point **of X**. A complete normed vector space is a **Banach space**.

COMPLETE SPACES

- $\mathbb{R}$. A Cauchy sequence is bounded, so it has a convergent subsequence (Bolzano–Weierstrass), so it converges by 4.2(3). This uses the completeness axiom.
- $\mathbb{R}^n$ with d_1, d_2 or d_∞. Each coordinate is Cauchy in $\mathbb{R}$, so it converges.
- Any **closed** subset of $\mathbb{R}^n$, such as $[0,1]$, $\mathbb{Z}$, or the Cantor set.
- Any **discrete** metric space. A Cauchy sequence (with $\varepsilon = ½$) is eventually constant.
- **$(C[a,b], d_\infty)$**: uniform limits of continuous functions are continuous. Proof below.
- ℓ^1, ℓ^2, ℓ^∞.
- Any compact metric space (Ch 5).

INCOMPLETE SPACES (AND THE CAUCHY SEQUENCE THAT SHOWS IT)

Space	Cauchy sequence with no limit in the space
$\mathbb{Q}$	Decimal truncations of $\sqrt{2}$: 1, 1.4, 1.41, … $\to \sqrt{2} \notin \mathbb{Q}$
$(0, 1)$	$1/n \to 0 \notin (0,1)$
$\mathbb{R} \setminus \{0\}$	$1/n$
$(C[0,1], d_1)$	Continuous ramps converging (in area) to a jump function (Q4 below)
Polynomials on $[0,1]$ with d_∞	Taylor partial sums of $e^t \to e^t$, which is not a polynomial
$\mathbb{R}$ with $d(x,y) = \lvert\arctan x - \arctan y\rvert$	$x_n = n$ is Cauchy ($\arctan n \to \pi/2$) but doesn't converge

COMPLETENESS IS NOT A TOPOLOGICAL PROPERTY

$\mathbb{R} \cong (0,1)$, but $\mathbb{R}$ is complete and $(0,1)$ isn't. The last row above is $\mathbb{R}$ with its usual topology but an incomplete metric. Completeness **is** preserved by Lipschitz-equivalent metrics, because they have the same Cauchy sequences.

THEOREM 4.4: CLOSED ⇔ COMPLETE (VERY EXAMINABLE)

1. If X is complete and $F \subseteq X$ is closed, then F is complete.
2. If $Y \subseteq X$ is complete (with the induced metric), then Y is closed in X, for **any** metric space X.

So inside a complete space: **a subset is complete $\Leftrightarrow$ it is closed**.

Proof. (1): a Cauchy sequence in F is Cauchy in X, so it converges to some $x \in X$. F is closed, so $x \in F$ (Theorem 2.11(4)). (2): let $y_n \in Y$ with $y_n \to x \in X$. Then (y_n) is convergent, hence Cauchy, so it converges in Y to some y. Limits are unique in X, so $x = y \in Y$. Hence Y is closed. ∎

Proof that (C[a,b], d_∞) is complete

Proof. Let (f_n) be Cauchy in d_∞. **Step 1, candidate limit:** for each t, $|f_n(t) - f_m(t)| \le d_\infty(f_n,f_m)$, so $(f_n(t))$ is Cauchy in $\mathbb{R}$ and converges. Call the limit $f(t)$. **Step 2, uniform convergence:** given ε, choose N with $|f_n(t) - f_m(t)| < \varepsilon$ for all t and all $m,n \ge N$. Let $m \to \infty$: $|f_n(t) - f(t)| \le \varepsilon$ for all t and all $n \ge N$, so $d_\infty(f_n, f) \to 0$. **Step 3, f is continuous (the "ε/3 argument"):** fix n with $\sup|f_n - f| < \varepsilon/3$. f_n is continuous at s, so choose δ with $|f_n(t) - f_n(s)| < \varepsilon/3$ when $|t - s| < \delta$. Then

$$|f(t) - f(s)| \le |f(t) - f_n(t)| + |f_n(t) - f_n(s)| + |f_n(s) - f(s)| < \varepsilon.$$

So $f \in C[a,b]$ and $f_n \to f$. ∎

GENERAL 3-STEP TEMPLATE FOR "PROVE X IS COMPLETE"

1. Take a Cauchy sequence and find a candidate limit (usually pointwise or coordinatewise).
2. Show the sequence converges to the candidate **in the metric**.
3. Show the candidate actually **belongs to** X.

4.3 The Contraction Mapping Theorem (Banach Fixed Point Theorem)

DEFINITION 4.5

$f : X \to X$ is a **contraction** if there is a constant k with $0 \le k < 1$ such that $d(f(x), f(y)) \le k \cdot d(x,y)$ for all x, y. A **fixed point** is an x^* with $f(x^*) = x^*$.

THEOREM 4.6: CONTRACTION MAPPING THEOREM

Let (X, d) be a **non-empty complete** metric space and let $f : X \to X$ be a contraction with constant k. Then:

1. f has a **unique** fixed point x^*;
2. for **any** starting point x_0, the iterates $x_{n+1} = f(x_n)$ converge to x^*;
3. error bound: $d(x_n, x^*) \le k^n/(1 - k) \cdot d(x_0, x_1)$.

Proof. **Step 1, successive distances shrink geometrically:** by induction, $d(x_n, x_{n+1}) \le k^n \, d(x_0, x_1)$. **Step 2, Cauchy:** for $m > n$, by the polygon inequality and a geometric series,

$$d(x_n, x_m) \le \Sigma_\{j=n\}\^\{m-1\} \, k^j \, d(x_0,x_1) \le k^n/(1-k) \cdot d(x_0,x_1) \to 0 \text{ as } n \to \infty$$

(because $k < 1$). **Step 3, convergence:** X is complete, so $x_n \to x^*$ for some $x^* \in X$. **Step 4, fixed point:** f is Lipschitz and hence continuous, so $f(x^*) = \lim f(x_n) = \lim x_{n+1} = x^*$. **Step 5, uniqueness:** if $f(x) = x$ and $f(y) = y$, then $d(x,y) = d(f(x),f(y)) \le k \cdot d(x,y)$. So $(1 - k)d(x,y) \le 0$, giving $d(x,y) = 0$. **Error bound:** let $m \to \infty$ in Step 2. ∎

EVERY HYPOTHESIS IS NEEDED

- **Completeness:** $f(x) = x/2$ on $(0, 1]$ is a contraction with no fixed point (0 is missing).
- **$k < 1$ strictly, with a fixed k:** $f(x) = x + 1/x$ on $[1, \infty)$ satisfies $|f(x) - f(y)| < |x - y|$ for $x \neq y$, but has **no** fixed point. ($|f(x) - f(y)| = |x-y|(1 - 1/(xy))$, and the factor can be arbitrarily close to 1.)
- **$k = 1$ isn't enough:** $f(x) = x + 1$ on $\mathbb{R}$ is an isometry with no fixed point.
- **f must map X into X:** always check this first in applications.

APPLICATION: SOLVING COS X = X

Let $X = [0, 1]$, which is closed in $\mathbb{R}$ and so complete, and $f(x) = \cos x$. (i) f maps X into $[\cos 1, 1] \subseteq [0,1]$. (ii) By the Mean Value Theorem, $|\cos x - \cos y| = |\sin \xi||x - y| \le \sin(1)\cdot|x - y|$, and $k = \sin 1 \approx 0.84 < 1$. So there is a unique solution in $[0,1]$ (≈ 0.739), and iterating cos from any starting point finds it.

Other standard applications: Picard–Lindelöf existence and uniqueness for ODEs $y' = F(t,y)$ (apply it to the integral operator on $C[a,b]$), integral equations, Newton's method convergence, and the inverse function theorem.

EXAM RECIPE FOR APPLYING THE THEOREM

1. Name a complete space X. Justify it, for example "closed subset of $\mathbb{R}$" or "$(C[a,b], d_\infty)$ is complete".
2. Show $f(X) \subseteq X$.
3. Find $k < 1$, usually via the MVT ($\sup|f'| < 1$) or a direct estimate.
4. Conclude there is a unique fixed point, with the iterates converging to it.

4.4 Further results (for harder questions)

THEOREM 4.7: CANTOR INTERSECTION THEOREM

Let X be complete and $F_1 \supseteq F_2 \supseteq \dots$ non-empty closed sets with $\operatorname{diam} F_n \to 0$. Then $\bigcap F_n$ is **exactly one point**. Conversely, if this holds for every such sequence, then X is complete.

Proof. Pick $x_n \in F_n$. For $m \ge n$, both x_m and x_n lie in F_n, so $d(x_m,x_n) \le \operatorname{diam} F_n \to 0$. The sequence is Cauchy, so $x_n \to x$. Each F_n is closed and contains the tail $x_n, x_{n+1}, \dots$, so $x \in F_n$ for every n. If y is also in the intersection, $d(x,y) \le \operatorname{diam} F_n \to 0$. ∎

HYPOTHESES NEEDED

Without closedness: $\bigcap(0, 1/n) = \varnothing$. Without diam $\to 0$: $\bigcap[n, \infty) = \varnothing$ in $\mathbb{R}$.

THEOREM 4.8: BAIRE CATEGORY THEOREM

In a complete metric space, a countable intersection of dense open sets is dense. Equivalently, a non-empty complete metric space is **not** a countable union of nowhere-dense sets. *Consequences:* $\mathbb{R}$ is uncountable (it isn't a countable union of singletons); $\mathbb{Q}$ is not complete under any metric equivalent to the usual one; there exist continuous nowhere-differentiable functions.

THEOREM 4.9: COMPLETION

Every metric space X sits isometrically as a **dense** subset of a complete metric space $\hat{X}$ (its **completion**), which is unique up to isometry. For example, the completion of $\mathbb{Q}$ is $\mathbb{R}$, and the completion of $(C[0,1], d_1)$ is $L^1[0,1]$.

4.5 Practice questions

EASY 1. Is $[0, \infty)$ complete? Is $\mathbb{Z}$? Is $\{1/n : n \in \mathbb{N}\}$?

Solution

Yes: it's closed in the complete space $\mathbb{R}$. Yes: it's closed, and Cauchy sequences in $\mathbb{Z}$ are eventually constant. No: $(1/n)$ is Cauchy, but its limit 0 isn't in the set (it isn't closed).

EASY 2. Show that $x_n = 1 + ½ + \ldots + 1/n$ is not Cauchy in $\mathbb{R}$.

Solution

x_{2n} − $x_n = 1/(n+1) + \ldots + 1/(2n) \geq n \cdot 1/(2n) = ½$ for every n. So with $\varepsilon = ½$, no N works. Note that $x_{n+1} - x_n \to 0$ is **not** enough for Cauchy.

MEDIUM 3. Show that $f(x) = ½(x + 2/x)$ is a contraction on $[1, 2]$. What is its fixed point?

Solution

$f'(x) = ½(1 - 2/x^2)$. On $[1,2]$, $2/x^2 \in [½, 2]$, so $f' \in [-½, ¼]$ and $|f'| \leq ½$. The minimum of f is at $\sqrt{2}$, where $f = \sqrt{2}$; also $f(1) = 1.5$ and $f(2) = 1.5$. So $f([1,2]) \subseteq [\sqrt{2}, 1.5] \subseteq [1,2]$. $[1,2]$ is complete, so there is a unique fixed point: $x = ½(x + 2/x) \Leftrightarrow x^2 = 2$, giving $x = \sqrt{2}$. This is the Babylonian / Newton method for $\sqrt{2}$. On $[1,2] \cap \mathbb{Q}$, f is still a contraction mapping $\mathbb{Q}$ to $\mathbb{Q}$ but has **no** fixed point, which shows completeness is needed.

HARD 4. Show that $(C[0,1], d_1)$ is not complete.

Solution

Define $f_n(t) = 0$ on $[0, ½]$, linear from 0 to 1 on $[½, ½ + 1/n]$, and 1 on $[½ + 1/n, 1]$ (for $n \geq 2$). For $m > n$, f_m and f_n differ only on $[½, ½ + 1/n]$, by at most 1, so $d_1(f_m, f_n) \leq 1/n$. The sequence is **Cauchy**.
Suppose $f_n \to f \in C[0,1]$ in d_1. On $[0,½]$: $\int_0^{½}|f| = \int_0^{½}|f - f_n| \leq d_1(f, f_n) \to 0$, so $f = 0$ on $[0,½]$ (by continuity). For any $c > ½$, once $½ + 1/n < c$, the same argument on $[c, 1]$ gives $f = 1$ there. Hence $f = 1$ on $(½, 1]$. Then f jumps at ½ and is not continuous, a contradiction.

HARD 5. Let X be complete and $f : X \to X$ with f^m (the m-th iterate) a contraction for some m. Show that f has a unique fixed point.

Solution

By the theorem, f^m has a unique fixed point x^*. Then $f^m(f(x^*)) = f(f^m(x^*)) = f(x^*)$, so $f(x^*)$ is also a fixed point of f^m. By uniqueness, $f(x^*) = x^*$. Any fixed point of f is a fixed point of f^m, so it is unique. (This is used in the ODE version of Picard's theorem on large intervals.)

Chapter 5: Compactness

⏱ 5:10–5:35 (25 min)

Learn the open-cover definition word for word, the three equivalent forms in metric spaces, Heine–Borel (and where it fails), and the theorems on continuous maps from compact spaces. Practise "show this is not compact" by finding a bad open cover or a bad sequence.

5.1 The idea

Intuition. Compactness is the topological version of "finite". Finite sets have nice properties: every function on them has a maximum, and every sequence in them repeats some value infinitely often. Compact sets keep those properties even when infinite. In $\mathbb{R}^n$, compact simply means **closed and bounded**. In general spaces you need the definition below.

5.2 Definitions

DEFINITION 5.1: OPEN COVER, COMPACT

- An **open cover** of $K \subseteq X$ is a collection $\{U_i\}_{i\in I}$ of open sets with $K \subseteq \bigcup U_i$.
- A **subcover** is a subcollection that still covers K. It is **finite** if it has finitely many sets.
- K is **compact** if **every** open cover of K has a **finite subcover**.

THE #1 MISREADING

"Has a finite open cover" is **always** true: $\{X\}$ is one. The definition says that **every** open cover can be reduced to finitely many sets. To prove compactness you must handle an **arbitrary** cover. To disprove it, you only need **one** cover with no finite subcover.

DEFINITION 5.2: SEQUENTIALLY COMPACT, TOTALLY BOUNDED

- X is **sequentially compact** if every sequence in X has a subsequence converging to a point **of X**.
- X is **totally bounded** if for every $\varepsilon > 0$, X is covered by **finitely many** balls of radius ε (an **ε-net**).

Totally bounded $\Rightarrow$ bounded, but **not conversely**. An infinite discrete space is bounded (diam = 1), but with $\varepsilon = \frac{1}{2}$ each ball is a single point, so infinitely many balls are needed.

5.3 The main equivalence (metric spaces)

THEOREM 5.3: FOR A METRIC SPACE X, THE FOLLOWING ARE EQUIVALENT

1. X is compact;
2. X is sequentially compact;
3. X is complete and totally bounded.

In general topological spaces, (1) and (2) can differ. In metric spaces you can use whichever is most convenient. Sequences are usually easiest.

Proof. **Sketch.** (1)⇒(2): if a sequence had no convergent subsequence, each point x would have a ball containing x_n for only finitely many n. Finitely many such balls cover X, so the sequence would have only finitely many terms, which is a contradiction. (2)⇒(3): a Cauchy sequence has a convergent subsequence, so it converges by 4.2(3). If X weren't totally bounded, you could choose points pairwise $\geq \varepsilon$ apart, and that sequence has no Cauchy (so no convergent) subsequence. (3)⇒(2): from any sequence, use finite ε-nets with $\varepsilon = 1, \tfrac{1}{2}, \tfrac{1}{3}, \ldots$ and a diagonal argument to extract a Cauchy subsequence. It converges by completeness. (2)⇒(1): use the Lebesgue number lemma (5.9) together with total boundedness. ∎

5.4 Heine–Borel and its limits

THEOREM 5.4: HEINE–BOREL

A subset of $\mathbb{R}^n$ (usual metric) is compact $\Leftrightarrow$ it is **closed and bounded**.

Proof. ⇒: Theorem 5.5. ⇐: a bounded sequence in $\mathbb{R}^n$ has a convergent subsequence (Bolzano–Weierstrass, coordinate by coordinate). Its limit lies in the set because the set is closed. So the set is sequentially compact. ∎

THEOREM 5.5: COMPACT ⇒ CLOSED AND BOUNDED (IN ANY METRIC SPACE)

Proof. **Bounded:** the balls $B(x_0, n)$, $n \in \mathbb{N}$, form an open cover. A finite subcover means $K \subseteq B(x_0, N)$. **Closed:** if $k_n \in K$ and $k_n \to x$, some subsequence converges in K. That subsequence also converges to x, so by uniqueness of limits $x \in K$. (In topological spaces: compact subsets of **Hausdorff** spaces are closed.) ∎

CLOSED + BOUNDED DOES NOT IMPLY COMPACT OUTSIDE $\mathbb{R}^n$

- **Infinite discrete space:** closed and bounded. The cover by singletons $\{\{x\} : x \in X\}$ has no finite subcover.
- **Closed unit ball of ℓ^2:** the vectors $e_n = (0,\dots,0,1,0,\dots)$ satisfy $d(e_n, e_m) = \sqrt{2}$ for $n \neq m$, so no subsequence is Cauchy.
- **Closed unit ball of $(C[0,1], d_\infty)$:** $f_n(t) = t^n$ has no uniformly convergent subsequence, because the pointwise limit is discontinuous.
- **$X = (0,1]$ with the usual metric:** X is closed in itself and bounded, but not compact. $1/n$ has no subsequence converging in X.
- **$\mathbb{R}$ with $d/(1+d)$:** bounded, closed in itself, not compact.

The right generalisation is **complete + totally bounded**.

5.5 Key theorems about compact sets

THEOREM 5.6: CLOSED SUBSETS OF COMPACT SETS ARE COMPACT

Proof. Let $F \subseteq K$ be closed and $\{U_i\}$ an open cover of F. Then $\{U_i\} \cup \{X \setminus F\}$ is an open cover of K. Take a finite subcover and drop $X \setminus F$; what remains still covers F. (Sequential version: a sequence in F has a subsequence converging in K, and its limit is in F because F is closed.) ∎

THEOREM 5.7: CONTINUOUS IMAGE OF COMPACT IS COMPACT

If $f : X \to Y$ is continuous and $K \subseteq X$ is compact, then $f(K)$ is compact.

Proof. Let $\{V_i\}$ be an open cover of $f(K)$. Then $\{f^{-1}(V_i)\}$ is an open cover of K, because each $f^{-1}(V_i)$ is open by continuity. A finite subcover $f^{-1}(V_1), \dots, f^{-1}(V_n)$ gives $f(K) \subseteq V_1 \cup \dots \cup V_n$. ∎

COROLLARY 5.8: EXTREME VALUE THEOREM

A continuous $f : K \to \mathbb{R}$ on a non-empty compact space is **bounded** and **attains** its max and min. Proof: $f(K)$ is a compact, hence closed and bounded, subset of $\mathbb{R}$, so it contains its sup and inf.

Failure without compactness: $f(x) = x$ on $(0,1)$ attains no max or min, and $1/x$ on $(0,1]$ is unbounded.

THEOREM 5.9: HEINE–CANTOR

A continuous function from a **compact** metric space to a metric space is **uniformly continuous**.

Proof. Suppose not. Then there are $\varepsilon > 0$ and x_n, y_n with $d(x_n, y_n) < 1/n$ but $d(f(x_n), f(y_n)) \geq \varepsilon$. By sequential compactness, some subsequence $x_{n_k} \to x$. Then $y_{n_k} \to x$ too, since $d(y_{n_k}, x) \leq 1/n_k + d(x_{n_k}, x)$. By continuity, $f(x_{n_k})$ and $f(y_{n_k})$ both converge to $f(x)$, so their distance tends to 0. That contradicts $\geq \varepsilon$. ∎

THEOREM 5.10: COMPACT → HAUSDORFF BIJECTIONS ARE HOMEOMORPHISMS

If $f : X \to Y$ is a continuous bijection, X is compact and Y is Hausdorff (for example, a metric space), then f is a homeomorphism.

Proof. We need f^{-1} to be continuous, that is, $(f^{-1})^{-1}(C) = f(C)$ closed for every closed $C \subseteq X$. A closed subset of a compact space is compact (5.6), its continuous image is compact (5.7), and compact subsets of a Hausdorff space are closed (5.5). ∎

COMPARE

This is why $[0,2\pi) \to S^1$ fails to be a homeomorphism: $[0,2\pi)$ is not compact. The same map on the compact space $[0,2\pi]$ isn't a bijection, but it passes to a homeomorphism once the endpoints are glued together.

THEOREM 5.11: LEBESGUE NUMBER LEMMA

If X is a compact metric space and $\{U_i\}$ is an open cover, there is a $\delta > 0$ (a **Lebesgue number**) such that every set of diameter $< \delta$ lies inside a single U_i.

THEOREM 5.12: FINITE INTERSECTION PROPERTY (FIP)

A collection of sets has the **FIP** if every finite subcollection has non-empty intersection. X is compact $\Leftrightarrow$ every collection of closed sets with the FIP has non-empty total intersection. Proof: take complements in the open-cover definition. *Consequence:* nested non-empty compact sets $K_1 \supseteq K_2 \supseteq \dots$ have non-empty intersection.

OTHER FACTS WORTH KNOWING

- Finite unions of compact sets are compact. Arbitrary intersections of compact sets in a Hausdorff space are compact.
- A compact metric space is complete, totally bounded, bounded and **separable**.
- Products: $X \times Y$ is compact $\Leftrightarrow$ X and Y are (Tychonoff's theorem extends this to arbitrary products).
- If K is compact, F is closed and $K \cap F = \varnothing$, then $d(K, F) > 0$. This can fail for two disjoint closed sets: the x-axis and the curve $y = 1/x$ are at distance 0.
- Compactness is a **topological** property.

5.6 Compactness in topological spaces

EXAMPLES

- Any finite space is compact, under any topology.
- Any indiscrete space is compact (the only open cover is {X}, possibly with ∅).
- Any set with the **cofinite** topology is compact. One open set U_0 covers everything except finitely many points, and finitely many more sets cover those.
- An infinite **discrete** space is **not** compact.

5.7 Practice questions

EASY 1. Show directly from the definition that (0, 1) is not compact.

Solution

The sets $U_n = (1/n, 1)$, $n \geq 2$, form an open cover. Any finite subcollection has union $(1/N, 1)$ for its largest index N, and this misses $1/(2N)$. So there is no finite subcover.

EASY 2. Which are compact? (a) $[0,1] \cup [2,3]$ (b) $\{1/n : n \in \mathbb{N}\} \cup \{0\}$ (c) $\mathbb{Z}$ (d) $\{(x,y) : x^2 + y^2 = 1\}$ (e) $\mathbb{Q} \cap [0,1]$

Solution

(a) Yes: closed and bounded. (b) Yes: closed (it contains its only limit point, 0) and bounded. (c) No: unbounded. (d) Yes: closed (preimage of {1}) and bounded. (e) No: not closed, since $\sqrt{2}/2$ is a limit of points in it but not in the set.

MEDIUM 3. Prove directly from the open-cover definition that $K = \{0\} \cup \{1/n : n \in \mathbb{N}\}$ is compact.

Solution

Let $\{U_i\}$ be any open cover. Some U_0 contains 0. It is open, so it contains $(-\varepsilon, \varepsilon)$ for some ε, and hence contains $1/n$ for all $n > 1/\varepsilon$. Only finitely many points $1, ½, \ldots, 1/N$ are left, and each lies in some U_i. Together that's $N + 1$ sets. This is the standard "the limit point swallows the tail" argument.

MEDIUM 4. Let K be compact and $f : K \to \mathbb{R}$ continuous with $f(x) > 0$ for all x. Show there's a $c > 0$ with $f(x) \geq c$ for all x.

Solution

By the EVT, f attains its minimum at some x_0. Let $c = f(x_0) > 0$. (It fails on $(0,1]$ with $f(x) = x$: the inf is 0 but not attained.)

HARD 5. Let (X, d) be compact and $f : X \to X$ satisfy $d(f(x), f(y)) < d(x,y)$ for all $x \neq y$. Prove that f has a unique fixed point. (Contrast with $x + 1/x$ on $[1,\infty)$ from Ch 4.)

Solution

Let $g(x) = d(x, f(x))$. It is continuous, and X is compact, so g attains its minimum at some x_0. If $f(x_0) \neq x_0$, then $g(f(x_0)) = d(f(x_0), f(f(x_0))) < d(x_0, f(x_0)) = g(x_0)$, which contradicts minimality. So $f(x_0) = x_0$. Uniqueness: if $x \neq y$ are both fixed, $d(x,y) = d(f(x),f(y)) < d(x,y)$, which is impossible. Compactness replaces the need for a uniform $k < 1$.

HARD 6. Show that an isometry $f : X \to X$ of a compact metric space into itself is surjective.

Solution

Suppose $a \notin f(X)$. $f(X)$ is compact, hence closed, so $\varepsilon = d(a, f(X)) > 0$. Let $x_n = f^n(a)$. For $m > n$, since f preserves distances, $d(x_n, x_m) = d(a, f^{m-n}(a)) \geq \varepsilon$, because $f^{m-n}(a) \in f(X)$. So (x_n) has no Cauchy subsequence, which contradicts sequential compactness.

Bonus: connectedness (if your course covers it)

Some courses put this in Chapter 5 or 6. It's short and often appears in exam questions about homeomorphisms, so it's worth skimming even if it isn't on your list.

DEFINITIONS

- X is **disconnected** if $X = U \cup V$ with U, V open, disjoint and non-empty (a **separation**). Otherwise X is **connected**. Equivalently, the only clopen sets are $\varnothing$ and X, or every continuous $f : X \to \{0,1\}$ is constant.
- X is **path-connected** if any two points are joined by a continuous path $\gamma : [0,1] \to X$.
- A **connected component** is a maximal connected subset.

KEY RESULTS

- The connected subsets of $\mathbb{R}$ are exactly the **intervals**.
- Continuous images of connected spaces are connected. This gives the **Intermediate Value Theorem**.
- Path-connected $\Rightarrow$ connected, but **not** conversely. The **topologist's sine curve** $\{(x, \sin(1/x)) : x > 0\} \cup (\{0\} \times [-1,1])$ is connected but not path-connected.
- Open connected subsets of $\mathbb{R}^n$ are path-connected.
- The closure of a connected set is connected. A union of connected sets sharing a common point is connected.
- $\mathbb{Q}$ is **totally disconnected**: $\{q < \sqrt{2}\}$ and $\{q > \sqrt{2}\}$ separate it. Discrete spaces with ≥ 2 points are disconnected.

USING CONNECTEDNESS TO SHOW $X \not\cong Y$

If $f : X \to Y$ is a homeomorphism, then $X \setminus \{p\} \cong Y \setminus \{f(p)\}$. Removing any point disconnects $\mathbb{R}$, but $\mathbb{R}^2$ minus a point stays connected, so $\mathbb{R} \not\cong \mathbb{R}^2$. Removing 0 from $[0,1)$ leaves it connected, but removing any point from $(0,1)$ disconnects it, so $[0,1) \not\cong (0,1)$. Similarly $S^1 \not\cong [0,1]$.

Proof-template toolkit

Most exam questions are one of these patterns. Learn the first move for each.

To prove…	Standard first move
d is a metric	Check M1, M2, M3 in order. For M3, add and subtract, or use an increasing subadditive φ.
d is **not** a metric	Give one numerical counterexample to one axiom.
U is open	(a) "Let $x \in U$. Take $r = \ldots$" and show $B(x,r) \subseteq U$; or (b) $U = f^{-1}(\text{open})$ for continuous f; or (c) U is a union or finite intersection of open sets.
F is closed	(a) The complement is open; (b) $F = f^{-1}(\text{closed})$; (c) sequences: $x_n \in F$, $x_n \to x \Rightarrow x \in F$.
A set is **not** open	Find one point of the set where every ball leaves the set.
A set is **not** closed	Find a sequence in the set converging to a point outside it.
$x \in \text{cl } A$	Find a sequence in A converging to x, or show every ball meets A.
f is continuous	ε–δ, or build from known continuous maps, or show it's Lipschitz.
f is **not** continuous	One sequence $x_n \to x$ with $f(x_n) \not\to f(x)$.
f is **not** uniformly continuous	Sequences x_n, y_n with $d(x_n,y_n) \to 0$ but $d(f(x_n),f(y_n)) \geq \varepsilon$.
$X \cong Y$	Write down an explicit bijection and check continuity both ways. Or use a continuous bijection from a compact to a Hausdorff space.
$X \ncong Y$	A topological property that differs: compact, connected, number of components after removing a point, Hausdorff, separable…
(x_n) is Cauchy	Bound $d(x_m,x_n)$ for $m > n$ by something $\to 0$ that depends only on n. A geometric series often works.
X is complete	Closed subset of a known complete space, or the 3-step template (candidate limit, convergence, membership).
X is **not** complete	One Cauchy sequence whose limit is missing.
A unique solution exists	The Contraction Mapping Theorem: complete space, maps into itself, $k < 1$.
K is compact	In $\mathbb{R}^n$: closed and bounded. Otherwise: sequential compactness, a closed subset of a compact set, a continuous image of a compact set, or an arbitrary open cover.

K is **not** compact	One open cover with no finite subcover, or one sequence with no convergent subsequence, or show it isn't closed or isn't bounded.
(X, τ) is not metrizable	Show it isn't Hausdorff.

Counterexample bank

False statement	Counterexample
Not open ⇏ closed	$[0,1)$, or $\mathbb{Q}$, in $\mathbb{R}$
An infinite intersection of open sets is open	$\bigcap(-1/n,\ 1/n) = \{0\}$
An infinite union of closed sets is closed	$\bigcup[1/n,\ 1] = (0,1]$
cl B(x,r) = $\bar{B}$(x,r)	Discrete metric with r = 1
cl(A ∩ B) = cl A ∩ cl B	A = $\mathbb{Q}$, B = $\mathbb{R} \setminus \mathbb{Q}$
Limits are unique (in every topological space)	Indiscrete or cofinite topology
Images of open/closed sets are open/closed	x^2 on $(-1,1)$; arctan on $\mathbb{R}$
Continuous ⇏ uniformly continuous	x^2 on $\mathbb{R}$; $1/x$ on $(0,1)$
Uniformly continuous ⇏ Lipschitz	$\sqrt{x}$ on $[0,1]$
A continuous bijection is a homeomorphism	$[0,2\pi) \to S^1$; identity $(\mathbb{R}, \text{discrete}) \to (\mathbb{R}, \text{usual})$
Boundedness/completeness are topological	$\mathbb{R} \cong (0,1)$
Topologically equivalent ⇏ Lipschitz equivalent	d and d/(1+d) on $\mathbb{R}$
d_1 and d_∞ on C[0,1] are equivalent	t^n: $d_1 \to 0$, d_∞ = 1
$x_{n+1} - x_n \to 0$ ⇏ Cauchy	Harmonic partial sums, or $\sqrt{n}$
Every Cauchy sequence converges	$1/n$ in $(0,1)$; $\sqrt{2}$ approximations in $\mathbb{Q}$
A weak contraction d(fx,fy) < d(x,y) has a fixed point	$x + 1/x$ on $[1,\infty)$ (true on **compact** spaces)
A contraction on any space has a fixed point	$x/2$ on $(0,1]$
Nested non-empty closed sets intersect	$[n, \infty)$ in $\mathbb{R}$ (true if compact, or complete with diam $\to 0$)
Closed + bounded ⇏ compact	Infinite discrete space; unit ball of ℓ^2; $(0,1]$ in itself
Bounded ⇏ totally bounded	Infinite discrete space
Disjoint closed sets are a positive distance apart	x-axis and $y = 1/x$
Connected ⇏ path-connected	Topologist's sine curve

Glossary of all terminology

Accumulation point / limit point / cluster point (of a set A)

A point x such that every ball around x contains a point of A other than x.

Banach space

A complete normed vector space.

Basis (for a topology)

A collection of open sets such that every open set is a union of members of the collection.

Bijection

A function that is injective and surjective, so it has an inverse.

Boundary ∂A

$\operatorname{cl} A \setminus \operatorname{int} A$: the points where every ball meets both A and its complement.

Bounded set

A set contained in some ball; equivalently, one with finite diameter.

Cauchy sequence

$\forall \varepsilon\ \exists N\ \forall m,n \geq N\colon d(x_m,x_n) < \varepsilon$.

Clopen

Both open and closed.

Closed ball $\bar{B}(x,r)$

$\{y : d(x,y) \leq r\}$.

Closed set

A set whose complement is open; equivalently, one containing the limits of all its convergent sequences.

Closure cl A

The smallest closed set containing A, which equals $A \cup A'$, which equals the set of limits of sequences in A.

Coarser / finer topology

$\tau_1 \subseteq \tau_2$ means τ_1 is coarser and τ_2 is finer.

Cofinite topology

The open sets are $\varnothing$ and the sets with finite complement.

Compact

Every open cover has a finite subcover.

Complete

Every Cauchy sequence converges in the space.

Completion

A complete space containing X isometrically as a dense subset.

Connected

Cannot be split into two disjoint non-empty open sets.

Continuous

The preimage of every open set is open; equivalently ε–δ, or the sequential condition.

Contraction

$d(f(x),f(y)) \le k \cdot d(x,y)$ for a fixed $k < 1$.

Convergent sequence

$d(x_n, x) \to 0$.

Dense

$\operatorname{cl} A = X$; every non-empty open set meets A.

Derived set A′

The set of limit points of A.

Diameter

$\sup\{d(a,b) : a,b \in A\}$.

Discrete metric / topology

$d(x,y) = 1$ for $x \neq y$; every subset is open.

Distance to a set d(x,A)

$\inf\{d(x,a) : a \in A\}$. It is $0 \Leftrightarrow x \in \operatorname{cl} A$.

Equivalent metrics (Lipschitz / strong)

$\alpha d \le d' \le \beta d$ for some $\alpha, \beta > 0$.

Equivalent metrics (topological)

The two metrics have the same open sets.

Exterior

$\operatorname{int}(X \setminus A)$.

Finite intersection property

Every finite subcollection has non-empty intersection.

Fixed point

A point x with $f(x) = x$.

Hausdorff (T_2)

Distinct points have disjoint open neighbourhoods.

Homeomorphism

A continuous bijection with a continuous inverse.

Image f(A)

$\{f(a) : a \in A\}$.

Indiscrete (trivial) topology

$\{\varnothing, X\}$.

Induced (subspace) metric / topology

The restriction of d to Y; the open sets are $U \cap Y$ with U open in X.

Interior int A

The largest open subset of A; the points with a ball inside A.

Isolated point

A point $a \in A$ with some ball meeting A only in a.

Isometry

A distance-preserving map.

Lebesgue number

A δ such that every set of diameter $< \delta$ lies in one member of the cover.

Lipschitz

$d(f(x),f(y)) \le L \cdot d(x,y)$.

Metric

A function satisfying M1 (positivity), M2 (symmetry) and M3 (triangle inequality).

Metrizable

A topology that comes from some metric.

Neighbourhood of x

A set containing an open set that contains x.

Norm

$\|v\| \ge 0$ with equality iff $v = 0$; $\|\lambda v\| = |\lambda|\|v\|$; triangle inequality.

Nowhere dense

$\operatorname{int}(\operatorname{cl} A) = \varnothing$.

Open ball B(x,r)

$\{y : d(x,y) < r\}$.

Open cover

A collection of open sets whose union contains the set.

Open set

Every point has a ball around it inside the set.

Path-connected

Any two points are joined by a continuous path.

Preimage $f^{-1}(B)$

$\{x : f(x) \in B\}$.

Pseudometric

Like a metric, but $d(x,y) = 0$ is allowed for $x \neq y$.

Separable

Has a countable dense subset.

Sequentially compact

Every sequence has a subsequence converging in the space.

Sphere S(x,r)

$\{y : d(x,y) = r\}$.

Subsequence

(x_{n_k}) with $n_1 < n_2 < \dots$.

Supremum / infimum

The least upper bound / greatest lower bound.

Topological property

A property preserved by homeomorphisms.

Topology

A collection of subsets containing $\varnothing$ and X, closed under arbitrary unions and finite intersections.

Totally bounded

For every ε, finitely many ε-balls cover the space.

Ultrametric

A metric with $d(x,z) \leq \max\{d(x,y), d(y,z)\}$.

Uniformly continuous

The ε–δ condition with δ independent of the point.

5:35 recap checklist

⏱ 5:35–5:45 (10 min)

Close this guide. On a blank page, write as many of these as you can from memory. Then tick off what you got and circle what you missed. The circled items are where to start next session.

Definitions to write out exactly

- ☐ Metric (M1–M3)
- ☐ Open set; closed set
- ☐ Interior, closure, boundary, limit point, dense
- ☐ Convergent sequence; Cauchy sequence
- ☐ Topology (T1–T3); Hausdorff
- ☐ Continuity (ε–δ and open-set versions); uniform continuity
- ☐ Homeomorphism
- ☐ Complete; contraction
- ☐ Compact (open covers); sequentially compact; totally bounded

Theorems to state (and sketch proofs)

- ☐ Open balls are open (the $r - d(x,y)$ trick)
- ☐ Unions and finite intersections of open sets are open, with a counterexample for infinite intersections
- ☐ The four equivalent forms of continuity
- ☐ Closed subsets of complete spaces are complete; complete subspaces are closed
- ☐ Contraction Mapping Theorem, all 5 steps
- ☐ Compact $\Leftrightarrow$ sequentially compact $\Leftrightarrow$ complete + totally bounded (metric spaces)
- ☐ Heine–Borel; a counterexample outside $\mathbb{R}^n$
- ☐ Continuous image of compact is compact $\Rightarrow$ EVT
- ☐ Heine–Cantor; compact $\rightarrow$ Hausdorff continuous bijection is a homeomorphism

Counterexamples to recall instantly

- ☐ A set that is neither open nor closed
- ☐ A continuous function that is not uniformly continuous
- ☐ A continuous bijection that is not a homeomorphism

☐ An incomplete space, with its Cauchy sequence

☐ A closed, bounded, non-compact set

> **NEXT UP**
>
> Break 5:45–6:00, then **linear systems 6:00–6:50** (separate guide to follow), then the wrap-up at 6:50 and a hard stop at **7:00pm**.